

Executive Summary: Microstructure Factor Zoo

15 L2 Features → 1-Minute AAPL Returns | MGMTMFE 412 — Market Frictions

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Intuition

Every second, the AAPL order book contains a snapshot of every visible buy and sell order at 10 price levels. That snapshot encodes information: the *balance* between buy-side and sell-side pressure, the *shape* of the book (thin top vs deep support), the *pace* of trades being executed, and the *recent drift* in the weighted mid-price.

If any of these signals predict where the price will be 60 seconds from now, a trader who can read them faster than the market can extract small but consistent profits — **on every second of every trading day**.

The question is not whether the signals exist. They do. The question is whether the profit from acting on them survives the cost of acting on them.

The 15 Features

We construct features in three families:

Order flow imbalance (OFI): The ratio of bid-side to ask-side volume at the best quote, measured instantly and with 5s and 30s rolling averages. A buy-side skew (more willing buyers than sellers at the best prices) predicts near-term price increases.

Book shape: How the order book is *structured* across price levels — whether quantity increases or decreases as you move away from the best quote, and whether the top of the book is thin relative to deeper levels. A thin book amplifies price impact; a steep book slope signals liquidity fragility.

Price and trade dynamics: The microprice (weighted fair-value mid between bid and ask), its recent drift, realized short-term volatility, the arrival rate of trades, the net direction of executed flow (buyers hitting ask vs sellers hitting bid), and the rate at which the best-level queue is being consumed.

Results

The linear model (Ridge regression) gives the honest answer: **IC = 0.045**, stable in both training and out-of-sample periods. An IC of 4.5% means the model predicts the direction of

the 1-minute return correctly about 52–53% of the time. That is real predictive content — just barely above the 50% random baseline.

Model	Train IC	Test IC	Interpretation
Ridge (OLS)	0.045	0.046	Real signal, weak and stable
LightGBM	0.501	0.025	Severe overfitting, no OOS value

The machine learning failure is itself a finding. LightGBM (500 trees) appears to achieve IC = 0.50 in-sample — 10× better than OLS. But on the out-of-sample test set, it collapses to IC = 0.02: worse than the linear model. This is textbook overfitting: with 414,000 training bars and essentially zero signal-to-noise in the label, gradient boosting memorizes noise patterns. Reliable HFT machine learning requires years of data and domain-constrained model architectures, not off-the-shelf ensembles.

Which Features Matter

Ablation analysis (removing one feature at a time from the best model) identifies the three dominant signals:

1. **Realized volatility:** High recent volatility predicts more volatile future returns — not the direction, but the magnitude. This is the classic volatility-clustering effect.
 2. **Signed trade imbalance:** Net buying pressure over the last 30 seconds predicts upward price continuation. This is the microstructure analogue of momentum.
 3. **30s OFI:** The medium-term book skew carries more directional information than the instantaneous snapshot — consistent with research showing that order book signals are more predictive when aggregated over time.
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The Friction That Ends the Story

The gross IC of 0.045 is real. But converting it into profit requires trading — and trading requires crossing the bid-ask spread. For AAPL in early 2024, the average half-spread is \$0.0046 per share. With \$50K notional (~270 shares), one round-trip costs \$2.50 in TC. The gross PnL per trade from a 0.045 IC model is roughly \$0.13 per trade. Transaction costs are **19× the gross signal**.

This is an even worse ratio than the SPY lead-lag strategy (9× for AAPL there). The reason: the factor zoo model fires at 1-second bars, where microprice movements in 60 seconds are much smaller than the spread on average. The signal is real but the cost of access is not.

The bid-ask spread is the universal gatekeeper. Whether the opportunity is a lead-lag correlation, an FOMC announcement overshoot, or a book-imbalance signal, the spread screens out all participants except those for whom it is a revenue source. The market is not inefficient — it is *efficiently protected by friction*.

Connection to Market Frictions

All three strategies in this project confirm the same economic fact from different angles:

- **Lead-lag:** SPY predicts AAPL with IC 0.024. TC is $9\times$ the gross signal.
- **FOMC fade:** Pre-announcement spread widening creates AND limits the opportunity.
- **Factor zoo:** 15 L2 features produce IC 0.045. TC is $19\times$ the gross signal.

The common thread is that market frictions — specifically, the bid-ask spread arising from adverse-selection between informed and uninformed traders — are not incidental costs. They are the mechanism by which the market allocates its inefficiencies to the participants best positioned to exploit them.